

The University Interscholastic League

Number Sense Test • HS District • 2026

Contestant's Number _____

Read directions carefully
before beginning test

DO NOT UNFOLD THIS SHEET
UNTIL TOLD TO BEGIN

Final		
2nd		
1st		
Score		
Initials		

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. ALL PROBLEMS ARE TO BE SOLVED MENTALLY. Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

STOP -- WAIT FOR SIGNAL!

- | | |
|--|---|
| <p>(1) $852 + 748 =$ _____</p> <p>(2) $1152 - 154 =$ _____</p> <p>(3) $2438 \div 9$ has a remainder of _____</p> <p>(4) $24.48 \div 6 =$ _____ (decimal)</p> <p>(5) $14 \times \frac{13}{15} =$ _____ (mixed number)</p> <p>(6) $\frac{1}{8} =$ _____ % (decimal)</p> <p>(7) $26^2 =$ _____</p> <p>(8) $2 + 4 + 6 + 8 + 10 + \dots + 28 + 30 =$ _____</p> <p>(9) The product of the proper divisors of 16 is _____</p> <p>*(10) $3197 + 2952 + 1011 =$ _____</p> <p>(11) $1 + 3 \times (6 - 10) \div (15 - 21) =$ _____</p> <p>(12) $124 \times 13 =$ _____</p> <p>(13) $222 \times \frac{2}{37} =$ _____</p> <p>(14) $66\frac{2}{3}\%$ of 891 is _____</p> <p>(15) $12^2 - 14^2 = 13 \times$ _____</p> <p>(16) $2096 \times 4 + 16 =$ _____</p> <p>(17) 2 grams = .08 oz. and 6 oz. = _____ grams</p> <p>(18) If 4 medals cost \$19.00, what will 10 medals cost?
\$ _____</p> | <p>(19) The sum of the LCM of 21 and 28 and the GCD of 21 and 28 is _____</p> <p>*(20) $298 \times 257 + 456 =$ _____</p> <p>(21) $\sqrt{676} =$ _____</p> <p>(22) $2 + 1 - 3 - 4 - 7 - 11 =$ _____</p> <p>(23) $4\frac{2}{3} \times 3\frac{3}{4} =$ _____ (mixed number)</p> <p>(24) The multiplicative inverse of 1.41666... is _____</p> <p>(25) If $4x - 5 = 9$, then $2x =$ _____</p> <p>(26) $0.6888\dots =$ _____ (proper fraction)</p> <p>(27) $0.125 - 0.375 - 0.625 =$ _____ (fraction)</p> <p>(28) $46 \times 49 =$ _____</p> <p>(29) The 12th term of the sequence 1, 4, 7, 10, ... is 34.
The 9th term is _____</p> <p>*(30) $\sqrt{192425} =$ _____</p> <p>(31) $1.2^2 \div 0.6^2 \times 0.3^2 =$ _____ (fraction)</p> <p>(32) 20% of 800 is 60% of _____ (mixed number)</p> <p>(33) The perimeter of a square is 104 cm. The area of the square is _____ cm²</p> <p>(34) $0.006 =$ _____ % (fraction)</p> <p>(35) $242_5 =$ _____ 10</p> |
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- (36) $(7^5 + 2^5 + 10) \div 9$ has a remainder of _____
- (37) How many 3-digit numbers greater than 140 can be made using the digits 1, 9, 2, and 4 without repetition? _____
- (38) If $\frac{8x-1}{8x+1} + \frac{8x+1}{8x-1} = 2 + \frac{B}{64x^2-1}$, then $B =$ _____
- (39) $\frac{27}{41} - \frac{7}{10} =$ _____
- *(40) 4 yards — 1 foot — 6 inch = _____ centimeters
- (41) $43 \times 63 =$ _____
- (42) $\{t, u, e, s, d, a, y\} \cap \{t, h, u, r, s, d, a, y\}$ contains how many distinct elements? _____
- (43) The 5-digit number 3285k is divisible by 6. Find k, $k > 0$. _____
- (44) $32826_{12} \div 8_{12}$ has a remainder of _____ ₁₂
- (45) The midpoint of a line segment with endpoints (2, 8) and $(-2, 5)$ is (x, y) . Find $y - x$. _____
- (46) Let $(2x - 3)(2x - 8) = ax^2 + bx + c$. Find $a + b + c$. _____
- (47) $\prod_{k=3}^{k=7} (1 - k) =$ _____
- (48) If $\frac{1}{3} + \frac{1}{6} + \frac{1}{10} + \dots + \frac{1}{n} = \frac{9}{11}$, then $n =$ _____
- (49) The hypotenuse of a 45° - 45° - 90° triangle is $\sqrt{162}$. The sum of the lengths of the other sides is _____
- *(50) $2324 \times 2526 - 2728 =$ _____
- (51) $2026_7 \times 3_7 =$ _____ ₇
- (52) $\frac{17}{18} = 0.abbb\dots$ and $b - a =$ _____
- (53) The sum of the digits of a 3-digit number is 10. How many such numbers exist? _____
- (54) A bag contains 10 blue balls and 6 red balls. If the first ball randomly drawn is red, what is the probability the second one is blue? _____
- (55) $(2 + 8i)(2 - 5i) = a + bi$ and $a + b =$ _____
- (56) P, Q, and R are the roots of $2x^3 + 8x^2 - 2x = 6$. Find $PQR - (PQ + PR + QR) + P + Q + R$. _____
- (57) If ${}_8C_5 = k \times ({}_8P_5)$, then $k =$ _____
- (58) The point (2, 8) is reflected across the line $y = 26 - x$ to the point (h, k) . Find $k - h$. _____
- (59) $7^3 - 5 =$ _____ ₇
- *(60) $\sqrt[3]{323282026} =$ _____
- (61) $\lfloor 2.8 \rfloor - \lceil -2.5 \rceil =$ _____
- (62) Let T_n be the n th triangular number. Find $T_2 - T_8$. _____
- (63) The period of $y = 8 - 2\sin(\frac{\pi x}{2})$ is _____
- (64) $302^3 =$ _____
- (65) $323 \times 328 =$ _____
- (66) If $xy = 2$ and $x + y = 8$, then $x^3 + y^3 =$ _____
- (67) $23^{25} \div 14$ has a remainder of _____
- (68) $K \times \frac{4}{37} = 48$ and $K =$ _____
- (69) $\begin{vmatrix} 2 & 0 & 3 \\ 0 & 2 & 8 \\ 8 & 2 & 3 \end{vmatrix} =$ _____
- *(70) $\sqrt[4]{232425262728} + 2025 =$ _____
- (71) $28 \times 23 = 7 \times$ _____
- (72) The positive slope of the tangent from (25, 0) to the circle $x^2 + y^2 = 49$ is _____
- (73) The minimum value of $y = 2x^2 - 3x + 1$ is _____
- (74) Let $f(x) = 2x + 8$. Find $f(f^{-1}(-3))$. _____
- (75) $0.2111\dots$ base 4 = _____ base 4 (fraction)
- (76) $31\frac{1}{4}\% =$ _____ (fraction)
- (77) $\int_2^3 (x+1) dx + \int_3^8 (x+1) dx =$ _____
- (78) The nonzero critical value of $y = 2x^3 - 3x^2$ is (a, b). Find $a - b$. _____
- (79) $\frac{5}{8}$ of 75% of 48 is _____
- *(80) A right cylinder is 28" high with a 6" diameter. Its volume is _____ cm^3

DO NOT DISTRIBUTE TO STUDENTS BEFORE OR DURING THE CONTEST

University Interscholastic League - Number Sense Answer Key HS • District • 2026

*number) $x - y$ means an integer between x and y inclusive

NOTE: If an answer is of the type like $\frac{2}{3}$ it cannot be written as a repeating decimal

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|----------------------|-----------------------|---|---|
| (1) 1,600 | (19) 91 | (36) 1 | (57) $\frac{1}{120}$ |
| (2) 998 | *(20) 73,190 — 80,894 | (37) 22 | (58) 6 |
| (3) 8 | (21) 26 | (38) 4 | (59) 662 |
| (4) 4.08 | (22) — 2 | (39) $-\frac{17}{410}$ | *(60) 653 — 720 |
| (5) $12\frac{2}{15}$ | (23) $17\frac{1}{2}$ | *(40) 305 — 336 | (61) 4 |
| (6) 12.5 | (24) $\frac{12}{17}$ | (41) 2,709 | (62) — 33 |
| (7) 676 | (25) 7 | (42) 6 | (63) 4 |
| (8) 240 | (26) $\frac{31}{45}$ | (43) 6 | (64) 27,543,608 |
| (9) 64 | (27) $-\frac{7}{8}$ | (44) 6 | (65) 105,944 |
| *(10) 6,802 — 7,518 | (28) 2,254 | (45) 6.5, $\frac{13}{2}$, $6\frac{1}{2}$ | (66) 464 |
| (11) 3 | (29) 25 | (46) 6 | (67) 9 |
| (12) 1,612 | *(30) 417 — 460 | (47) — 720 | (68) 444 |
| (13) 12 | (31) $\frac{9}{25}$ | (48) 55 | (69) — 68 |
| (14) 594 | (32) $266\frac{2}{3}$ | (49) 18 | *(70) 2,584 — 2,855 |
| (15) — 4 | (33) 676 | *(50) 5,574,312 — 6,161,080 | (71) 92 |
| (16) 8,400 | (34) $\frac{3}{5}$ | (51) 6114 | (72) $\frac{7}{24}$ |
| (17) 150 | (35) 72 | (52) — 5 | (73) — .125, — $\frac{1}{8}$ |
| (18) 47.50 | | (53) 54 | (74) — 3 |
| | | (54) $\frac{2}{3}$ | (75) $\frac{13}{30}$ |
| | | (55) 50 | (76) $\frac{5}{16}$ |
| | | (56) 0 | (77) 36 |
| | | | (78) 2 |
| | | | (79) 22.5, $\frac{45}{2}$, $22\frac{1}{2}$ |
| | | | *(80) 12,325 — 13,621 |